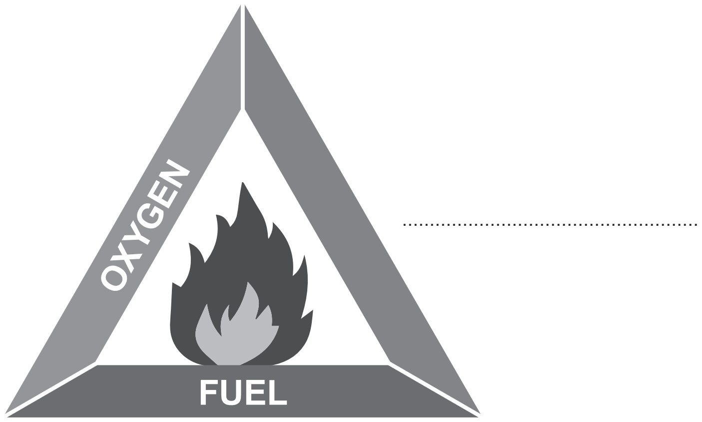
Examiner only

2. (a) In the diagram of the fire triangle, two sides have been labelled.



(i) **Complete the fire triangle** by labelling the third side. [1]

(ii) Complete the following sentences to explain how a fire is extinguished by the following methods. [2]

Using a fire blanket removes the .... from the fire triangle.

Cutting down trees in a forest fire removes the .... from the fire triangle.

(b) Choose a **number** from the box below to balance the equation for the burning of propane gas. [1]

2	**3**	**4**	**8**

$$C_3H_8 \quad + \quad 5O_2 \longrightarrow 3CO_2 \quad + \quad \boxed{}\, H_2O$$

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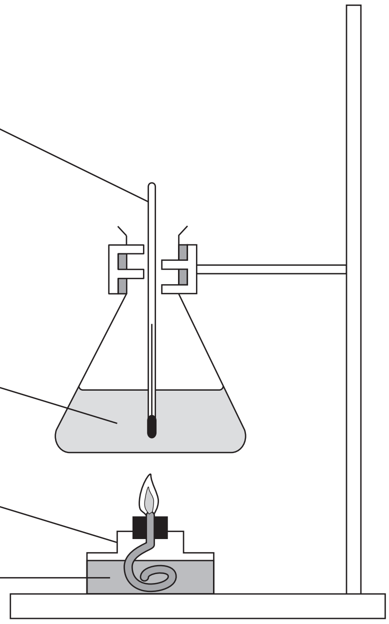
(c) A class used the apparatus shown to compare the combustion of different alcohols. They burned each alcohol for 1 minute. They measured the temperature of the water before and after burning each alcohol.



The table shows the increase in temperature of the water for each alcohol.

Alcohol	Temperature increase (°C)
methanol	8
ethanol	19
propanol	23
butanol	38

(i) The starting temperature of the water each time was 18 °C. Calculate the final temperature of the water after burning ethanol. [1]

Final temperature = ... °C

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(ii) Tick (✓) the question that the class were trying to answer. [1]

Which alcohol gives out the most heat energy?	
Which gases are produced when alcohols burn?	
Which alcohol has the lowest boiling point?	
Which alcohol burns for the longest?	

(d) Methanol has the chemical formula CH_3OH.

Calculate the relative molecular mass, M_r, of methanol. [2]

$A_r(C) = 12$ $A_r(H) = 1$ $A_r(O) = 16$

M_r = ...

8